LERATO THANDO MBHELE

Final-Year BSc Computer Science & Information Systems Student | Software Development

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PROFESSIONAL SUMMARY

Final-year BSc Computer Science and Information Systems student at the University of KwaZulu-Natal (UKZN), expected to graduate on 19 December 2026. Experience includes developing a multi-phase business management system (Windows Forms .NET desktop and web versions) as a final-year group project, concurrent systems programming in C++, machine-learning projects in Python, and working as a Demonstrator for COMP 100 (Introduction to Programming) for first-year students. Seeking graduate programms and junior software development or IT support roles.

EDUCATION

BSc Computer Science and Information Systems

University of KwaZulu-Natal (UKZN) • 2022 – Present (19 Dec 2026)

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Computer Systems & Operating Systems, Web Development, Systems Analysis & Design

National Senior Certificate (Grade 12)

Sizophumelela High School • Completed 2020

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, SQL

Web & Full-Stack: CSS, HTML, JavaScript, ASP.NET Framework,

Databases: SQL Server, MySQL, Database Design

Systems & Security: Concurrency & Multithreading, Memory Management,

Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib, Data Preprocessing, Model Training & Evaluation

Tools & Platforms: Git, GitHub, VS Code, Visual Studio

Operating Systems: Windows, Linux (Ubuntu/Zorin OS)

PROJECTS

Thywill Electrical Services — Job & Inventory Management System | C# / .NET / SQL Server

Final-Year Group Project | C# · Windows Forms · .NET Framework · SQL Server

Developed as part of a two-semester university group project to computerize job and resource management for an electrical and construction services business.

- Contributed to the Windows Forms desktop application, working with C#, business-service and data-access layers to support operational workflows.
- Worked on database-backed CRUD operations and data validation for business records, including customers, jobs, staff, and inventory.
- Implemented and maintained job-management workflows involving technician assignment, job status tracking, inventory requirements, and stock availability.

- Worked with service-cost calculations, 15% VAT totals and invoice generation, including PDF storage and attempted email delivery.

- Extended the project into a second-semester ASP.NET Web Forms application using C# and .NET Framework, applying a layered architecture to separate presentation, business logic and database access.

Project limitation: The university-controlled SQL Server database restricted independent database administration and testing. Database-dependent functionality was therefore subject to the university environment and its access constraints.

Student Performance Prediction System

Machine Learning Application | Solo Project

- Developed a Python machine-learning application to predict student academic performance using study habits, attendance and lifestyle-related features.

- Prepared and preprocessed the dataset, performed feature engineering, and trained classification models using Scikit-learn.

- Evaluated model performance using accuracy, precision, recall and F1-score, and visualized evaluation results using Matplotlib.

Technologies: Python, Scikit-learn, Pandas, Matplotlib

Concurrent Inventory & Trading System

C++ Multithreading | Group Project

- Contributed to a concurrent order-processing system using `std::thread` to process 20 orders across 9 warehouses.

- Implemented mutex-based synchronization using `std::mutex` to coordinate access to shared inventory data and help prevent race conditions and invalid stock updates.

- Used C++ smart pointers, including `std::shared_ptr`, to manage shared object ownership and reduce manual memory-management risks.

Technologies: C++, C++ Standard Library (STL)

CORE COMPETENCIES

Problem-Solving • Self-Directed Learning • Team Collaboration • Attention to Detail • Time Management • Adaptability

ADDITIONAL INFORMATION

- Interested in cloud infrastructure and systems-level software engineering.

English (proficient)